Shelter (SHL)
Interface Control Document

Version F

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Long Wavelength Array Project

Change Record

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Section 1

Description

1.1 Purpose

This document describes the interface control for the Shelter (SHL) subsystem of the Long Wavelength Array (LWA). The SHL subsystem is responsible for monitoring and controlling the environmental conditions, power distribution, and safety systems within the station shelter.

1.2 Scope

This Interface Control Document (ICD) defines:

- Physical interfaces to and from SHL
- Monitor and control (M&C) interfaces
- Safety interfaces
- Command and response message formats
- Management Information Base (MIB) entries

1.3 System Overview

The SHL subsystem performs the following functions:

- **Power Distribution**: Control and monitoring of rack-mounted PDUs and UPSs
- **Environmental Control**: Temperature monitoring and HVAC system interface
- **Weather Monitoring**: Integration with on-site weather station
- **Lightning Detection**: Interface with electric field mill for lightning alerts
- **Voltage Monitoring**: Detection of power flickers and outages
- **Safety Monitoring**: Smoke detection, water detection, and door status

Site-Specific Configuration

The specific hardware configuration varies by LWA site. The number of PDUs, type of HVAC controller, and availability of environmental sensors depends on the site installation. Site-specific configurations are defined in the `defaults.json` configuration file.

1.4 Related Documents

The following documents are related to this ICD:

- MCS Common ICD (LWA Engineering Memo MCS0005)
- MCS-NDP Interface Control Document
- MCS-ASP Interface Control Document

1.5 Order of Precedence

In the event of a conflict between this document and other project documentation, the MCS common ICD takes precedence, followed by this subsystem-specific ICD.

1.6 Document Organization

This document is organized as follows:

- **Section 1** (this section): General description and scope
- **Section 2**: Abbreviations and definitions
- **Section 3**: Physical interface specifications
- **Section 4**: Monitor and control interface
- **Section 5**: Safety interface
- **Section 6**: References
- **Appendix A**: Supported hardware reference

Section 2

Abbreviations and Definitions

2.1 Abbreviations

Abbreviation	Definition
APC	American Power Conversion (PDU/UPS manufacturer)
ECS	Environmental Control System
EFM	Electric Field Mill (lightning detector)
EnviroMux	Network Technologies environmental monitoring system
HVAC	Heating, Ventilation, and Air Conditioning
ICD	Interface Control Document
IceQube	IceQube V Series air conditioner
LWA	Long Wavelength Array
M&C	Monitor and Control
MC4002	Bard MC4002 HVAC controller
MCS	Monitor and Control System
MIB	Management Information Base
MJD	Modified Julian Date
MPM	Milliseconds Past Midnight
PDU	Power Distribution Unit
SHL	Shelter subsystem
SNMP	Simple Network Management Protocol
UDP	User Datagram Protocol
UPS	Uninterruptible Power Supply
VAC	Volts Alternating Current

2.2 Definitions

Rack A standard 19-inch equipment rack within the shelter. Each rack may have one or more PDUs.

PDU A Power Distribution Unit that provides switched power outlets for equipment. Controlled via SNMP.

UPS An Uninterruptible Power Supply that provides battery backup and power conditioning. Controlled via SNMP.

Set Point The target temperature for the HVAC system.

Differential The temperature offset from the set point at which cooling engages. For example, with a set point of 72°F and a differential of 2°F, cooling starts at 74°F.

SCRAM An emergency shutdown mode that immediately stops all monitoring threads but does not power off equipment.

Monitor Period The polling interval for a particular monitoring thread (e.g., temperature, PDU status).

Section 3

Physical Interfaces

3.1 Overview

The SHL subsystem interfaces with various hardware components through network connections. All monitored and controlled devices communicate via Ethernet using either SNMP or HTTP protocols.

3.2 Network Interfaces

3.2.1 Device Network Interfaces

All monitored devices are connected to the shelter's internal network (typically 172.16.1.x subnet), as summarized in Table 3.1.

Device Type	Protocol	Port	Notes
PDUs	SNMP	161	Various manufacturers
UPSs	SNMP	161	APC, TrippLite
Thermometers	SNMP	161	HWg STE, Comet P8510
EnviroMux	SNMP	161	Network Technologies
HVAC Controllers	HTTP	80	Bard MC4002, IceQube
WebRelay	HTTP	80	Control Relay Solutions

Table 3.1: Device Network Interfaces

3.2.2 External Software Interfaces

New in Version F

The SHL subsystem receives data from two external software processes via UDP multicast (Table 3.2): the lightning detector (Boltek EFM-100C electric field mill) and the voltage monitor (AC line quality monitoring). See Section 4 for details.

Interface	Multicast Address	Port	Source
Lightning Detector	224.168.2.9	7163	spinningCan.py
Voltage Monitor	224.168.2.10	7165	lineMonitor.py

Table 3.2: External Software UDP Multicast Interfaces

3.3 Power Interfaces

Site-Specific Configuration

The number of racks and PDUs varies by site. The configuration is defined in the site's `defaults.json` file under the `pdus.devices` section. Each device entry specifies the PDU type, IP address, and number of outlets.

3.3.1 Supported PDU Types

The following PDU types are supported:

- **APC**: APC AP7921B Switched Rack PDU (8 outlets)
- **Raritan**: Raritan PX2 Series (8 outlets)
- **Dominion**: Raritan Dominion PX (variable outlets)
- **TrippLite**: TrippLite PDUMH20ATNET (8 outlets)

3.3.2 Supported UPS Types

The following UPS types are supported:

- **APCUPS**: APC Smart-UPS with network management card
- **TrippLiteUPS**: TrippLite SmartOnline UPS

3.4 Environmental Interfaces

3.4.1 Temperature Monitoring

Temperature sensors are polled via SNMP. Two thermometer types are supported:

- **HWg STE**: HW group STE Ethernet thermometer (up to 2 sensors)
- **Comet**: Comet P8510 Ethernet thermometer (1 sensor)

3.4.2 Environmental Monitoring System

New in Version F

The Network Technologies E-MINI-LXO EnviroMux system provides integrated monitoring for:

- Temperature sensors (up to 2)
- Smoke detector
- Water/leak sensor
- Door open/closed sensor
- Airflow sensors

Site-Specific Configuration

Not all sites have an EnviroMux installed. When not present, the corresponding MIB entries (SMOKE, WATER, DOOR) will return error messages.

3.4.3 Weather Station

Weather data is obtained from a wview database (SQLite format). The weather station hardware is typically a Davis VantagePro2 with WeatherLink data logger.

Database location: `/var/wview/archive/wview-archive.sdb` (may vary by site)

Section 4

Monitor and Control Interface

4.1 Overview

The SHL subsystem communicates with the Monitor and Control System (MCS) via UDP messages. Commands are received on port 5008 and responses are sent to port 5009.

Site-Specific Configuration

The MCS communication ports are configurable via `mcs.message_in_port` and `mcs.message_out_port` in the site configuration file. The MCS host address is configurable via `mcs.message_host`.

4.2 Message Format

4.2.1 Command Message Format

Commands from MCS to SHL follow the standard LWA message format, as shown in Table 4.1.

Table 4.1: Command Message Format

Field	Bytes	Position	Description
Destination	3	0–2	Subsystem ID ("SHL")
Sender	3	3–5	Sender ID ("MCS")
Command	3	6–8	Command code
Reference	9	9–17	Reference number
Data Length	4	18–21	Length of data section
MJD	6	22–27	Modified Julian Date
MPM	9	28–36	Milliseconds Past Midnight
-	1	37	Space
Data	Variable	38+	Command-specific data

4.2.2 Response Message Format

Responses from SHL to MCS include the command status and any requested data (Table 4.2).

Table 4.2: Response Message Format

Field	Bytes	Position	Description
Destination	3	0–2	Destination ID
Sender	3	3–5	"SHL"
Command	3	6–8	Echo of command
Reference	9	9–17	Echo of reference number
Data Length	4	18–21	Length of data section
MJD	6	22–27	Response MJD
MPM	9	28–36	Response MPM
-	1	37	Space
Status	1	38	"A" (accept) or "R" (reject)
Summary	7	39–45	System status summary
Data	Variable	46+	Response data

4.3 Management Information Base (MIB)

The following MIB entries can be queried using the `RPT` command. All MIB values are returned as ASCII text strings.

4.3.1 General Information

General system information is available via the MIB entries listed in Table 4.3.

Table 4.3: General MIB Entries

MIB Entry	Description
`SUMMARY`	Current system status (7 characters)
`INFO`	Detailed status information (up to 256 characters)
`LASTLOG`	Last log message (up to 256 characters)
`SUBSYSTEM`	Subsystem identifier ("SHL")
`SERIALNO`	Subsystem serial number (site-specific, e.g., "SHL01")
`VERSION`	Software version

4.3.2 Power Distribution

The SHL subsystem monitors and controls power distribution equipment including PDUs and UPSs (Table 4.4). Each device is assigned to a rack number for addressing purposes.

Site-Specific Configuration

The number of racks and their configurations vary by site. For example:

- LWA1: 7 racks (mix of APC, Raritan, TrippLite PDUs/UPSs)
- LWA-SV: Similar to LWA1 (multiple racks, mix of PDU types)
- LWA-NA: 3 racks (APC PDUs and UPS)

Table 4.4: Power Distribution MIB Entries

MIB Entry	Description
`PORTS-AVAILABLE-R{n}`	Number of outlets available on rack {n}
`CURRENT-R{n}`	Total current draw (Amps) for rack {n}
`VOLTAGE-R{n}`	Input voltage (VAC) for rack {n}
`FREQUENCY-R{n}`	Input frequency (Hz) for rack {n}
`PWR-R{n}-{m}`	Power state of outlet {m} on rack {n} ("ON", "OFF", or "UNK")
`BATCHARGE-R{n}`	Battery charge percentage (UPS only)
`BATSTATUS-R{n}`	Battery status (UPS only)
`OUTSOURCE-R{n}`	Output power source (UPS only)

Where {n} is the rack number (1-indexed) and {m} is the outlet number (1-indexed).

The `BATSTATUS-R{n}` entries return one of: "Unknown", "Normal", "Low", or "Depleted".

The `OUTSOURCE-R{n}` entries return one of: "Other", "None", "Normal", "Bypass", "Battery", "Booster", or "Reducer".

4.3.3 Environmental Control System

The environmental control system maintains shelter temperature within acceptable limits (Table 4.5). The system supports two types of HVAC controllers:

- **Bard MC4002**: Multi-stage cooling/heating with lead/lag control, local and remote sensors, and multiple alarm types
- **IceQube V Series**: Single-stage cooling with lead/lag operation, enclosure/condenser temperature monitoring, and filter status

Table 4.5: Environmental Control System MIB Entries

MIB Entry	Description
`SET-POINT`	Current temperature set point (°F)
`DIFFERENTIAL`	Current cooling differential (°F)
`TEMPERATURE`	Mean shelter temperature (°F)
`TEMPERATURE-S{n}`	Individual sensor {n} temperature (°F)

New in Version F

The `TEMPERATURE-S{n}` entries provide access to individual temperature sensors. The sensor number is 1-indexed and includes sensors from all configured thermometers and EnviroMux devices.

The `TEMPERATURE` entry returns a "mean-max" temperature: if the spread between the highest and lowest sensor readings is $\leq$ 10°F, the mean is returned; otherwise, the maximum is returned. This prevents a single cold sensor from masking an overheating condition.

HVAC Controller Note

Not all HVAC capabilities are exposed through the MCS interface. Direct access to the HVAC controller's web interface provides additional functionality. See Section 4.6 for details.

4.3.4 Weather Station

Weather data is obtained by reading from a wview SQLite database (Table 4.6). The database is populated by the wview daemon which communicates with the weather station hardware.

Table 4.6: Weather Station MIB Entries

MIB Entry	Description
`WX-UPDATED`	Last update timestamp (YYYY-MM-DD HH:MM:SS)
`WX-TEMPERATURE`	Outside temperature (°F)
`WX-HUMIDITY`	Outside humidity (%)
`WX-PRESSURE`	Barometric pressure (inHg)
`WX-WIND`	Wind speed and direction
`WX-GUST`	Wind gust speed and direction
`WX-RAINFALL-RATE`	Current rainfall rate (in/hr)
`WX-RAINFALL-TOTAL`	Total rainfall accumulation (in)

The `WX-WIND` and `WX-GUST` entries return a formatted string: `{speed} mph at {direction} degrees` (e.g., `5.2 mph at 180 degrees`).

4.3.5 Lightning Detection

Lightning detection is provided by a Boltek EFM-100C Electric Field Mill (Table 4.7). The `spinningCan.py` process reads data from the EFM-100C and broadcasts lightning strike notifications via UDP multicast. The SHL subsystem subscribes to these notifications and maintains a rolling window of recent strikes.

Table 4.7: Lightning Detection MIB Entries

MIB Entry	Description
`LIGHTNING-RADIUS`	Detection radius (km), currently fixed at 15 km
`LIGHTNING-10MIN`	Strike count within radius in last 10 minutes
`LIGHTNING-30MIN`	Strike count within radius in last 30 minutes

4.3.6 Power Monitoring

New in Version F

The voltage monitor detects power flickers and outages on the 120V and 240V AC lines (Table 4.8). An Arduino-based voltage monitor board runs the `lineMonitor.py` process which broadcasts power events via UDP multicast.

Table 4.8: Power Monitoring MIB Entries

MIB Entry	Description
`POWER-FLICKER`	Power flicker detected (True/False)
`POWER-OUTAGE`	Power outage detected (True/False)

The voltage monitor tracks both 120V and 240V AC lines. A flicker is a brief voltage drop, while an outage is a sustained loss of power. Flicker status automatically clears after 5 minutes. Outage status clears when power is restored.

4.3.7 Environmental Sensors

New in Version F

Available through a correctly configured EnviroMux. See Table 4.9 for the available MIB entries.

Table 4.9: Environmental Sensors MIB Entries

MIB Entry	Description
`SMOKE`	Smoke detector status (True = smoke detected)
`WATER`	Water sensor status (True = water detected)
`DOOR`	Door status (OPEN or CLOSED)

Site-Specific Configuration

These MIB entries require an EnviroMux system to be configured. If no EnviroMux is present, querying these entries will return an error message indicating "No environmental monitoring system configured."

4.4 Control Commands

4.4.1 PNG — Ping

The `PNG` command tests connectivity with the SHL subsystem (Table 4.10).

Table 4.10: PNG Command

Field	Value
Command	`PNG`
Data	(none)
Response	Status only

4.4.2 INI — Initialize

The `INI` command initializes the SHL subsystem (Table 4.11). This starts all monitoring threads and sets initial HVAC parameters.

Table 4.11: INI Command

Field	Value
Command	`INI`
Data	{`setpoint`}&{`differential`}&{`rack_flags`}
Response	Status; on failure: exit code and message

The data field contains three ampersand-separated values:

- {`setpoint`}: Temperature set point in °F (e.g., `72`)
- {`differential`}: Cooling differential in °F (e.g., `1.0`)
- {`rack_flags`}: String of 1s and 0s indicating which racks are present (e.g., `1111111`)

Example: `72&1.0&1111111`

Exit Codes

`0x00`	Process accepted without error
`0x01`	Invalid temperature set point
`0x02`	Invalid temperature differential
`0x06`	Invalid command arguments
`0x07`	Blocking operation in progress
`0x08`	Subsystem already initialized

4.4.3 SHT — Shutdown

The `SHT` command shuts down the SHL subsystem (Table 4.12). This stops all monitoring threads.

Table 4.12: SHT Command

Field	Value
Command	`SHT`
Data	Mode (optional): "SCRAM", "RESTART", or "SCRAM RESTART"
Response	Status; on failure: exit code and message

Exit Codes

`0x00`	Process accepted without error
`0x06`	Invalid command arguments (unknown mode)
`0x07`	Blocking operation in progress

4.4.4 TMP — Set Temperature

The `TMP` command sets the HVAC temperature set point (Table 4.13).

Table 4.13: TMP Command

Field	Value
Command	`TMP`
Data	Temperature set point in °F
Response	Status; on failure: exit code and message

New in Version F

The `TMP` command now communicates with the HVAC controller to change the actual set point. Previously, this command only updated internal state and did not affect the HVAC system.

HVAC Controller Note

Valid temperature range is typically 60–110°F, but depends on site configuration.

Exit Codes

`0x00` Process accepted without error
`0x01` Invalid temperature set point
`0x09` Subsystem needs to be initialized

4.4.5 DIF — Set Differential

The `DIF` command sets the HVAC cooling differential (Table 4.14), the offset from set point at which cooling engages.

Table 4.14: DIF Command

Field	Value
Command	`DIF`
Data	Differential in °F
Response	Status; on failure: exit code and message

New in Version F

The `DIF` command now communicates with the HVAC controller to change the actual cooling offset. Previously, this command only updated internal state and did not affect the HVAC system.

Exit Codes

`0x00` Process accepted without error
`0x02` Invalid temperature differential
`0x09` Subsystem needs to be initialized

4.4.6 PWR — Power Control

The `PWR` command controls the power state of a PDU outlet (Table 4.15).

Table 4.15: PWR Command

Field	Value
Command	`PWR`
Data	{`rack`}{`port`}{`control`}
Response	Status; on failure: exit code and message

The data field is a concatenation of:

- {`rack`}: Single digit rack number (1–9)
- {`port`}: Two-digit port number (01–99)
- {`control`}: Three-character control word ("ON ", "OFF", or "CYC")

Example: `101ON` (turn on rack 1, port 1). Note the trailing space in the "ON " control word.

The "CYC" control word power cycles the outlet (off, delay, on).

Exit Codes

`0x00`	Process accepted without error
`0x03`	Invalid PDU rack number
`0x04`	Invalid PDU port number
`0x05`	Invalid PDU control keyword
`0x09`	Subsystem needs to be initialized

4.4.7 RPT — Report MIB Entry

The `RPT` command queries a MIB entry (Table 4.16).

Table 4.16: RPT Command

Field	Value
Command	`RPT`
Data	MIB entry name
Response	MIB value; on failure: error message

Example: `RPT TEMPERATURE` returns the mean shelter temperature.

4.5 System States

The SHL subsystem reports one of the status values listed in Table 4.17.

Table 4.17: System Status Values

Status	Description
`SHUTDWN`	System is shut down or not initialized
`BOOTING`	System is initializing
`NORMAL`	System is operating normally
`WARNING`	Warning condition (e.g., temperature approaching limit)
`ERROR`	Error condition requiring attention

4.6 HVAC Control Capabilities

New in Version F

This section documents the full set of HVAC controller capabilities. The set point and cooling offset are exposed through the `TMP` and `DIF` commands, respectively. The remaining capabilities are accessible through the HVAC controller's web interface.

4.6.1 Bard MC4002 Controller

The Bard MC4002 is an industrial HVAC controller accessible via HTTP at port 80. Its available functions are listed in Table 4.18.

Available Functions

Table 4.18: Bard MC4002 Functions

Function	Description
Status	Cooling/heating mode, stage active, alarm state
Alarms	List of active alarms (fire/smoke, temperature, controller)
Lead/Lag	Which unit (1 or 2) is currently lead
Temperatures	Average, local, and remote sensor readings
Set Point	Target temperature (exposed via `TMP`)
Cooling Offset	Differential (exposed via `DIF`)

Alarm Types

- Power loss 1 / Power loss 2
- Fire/smoke
- Low temperature
- High temperature 1 / High temperature 2
- Controller failure

4.6.2 IceQube Controller

The IceQube V Series controller is accessible via HTTP at port 80. Its available functions are listed in Table 4.19.

Available Functions

Table 4.19: IceQube Functions

Function	Description
Status	Cooling, heating, alarm, and filter status LEDs
Lead Status	Whether this unit is the lead unit
Temperatures	Enclosure and condenser temperatures
Set Point	Cooling on temperature (exposed via `TMP`)
Cooling Offset	Offset from set point (exposed via `DIF`)

4.6.3 WebRelay Control

Some sites use a WebRelay-10 device to control HVAC unit power for recovery from hung controllers. The available functions are listed in Table 4.20.

Available Functions

Table 4.20: WebRelay Functions

Function	Description
Get State	Query relay state (on/off)
Set State	Set relay to off (0), on (1), or cycle (2)

Note

WebRelay control is not exposed through MCS. It is available programmatically through the `shlQube` module functions: `get_webrelay_status()`, `get_webrelay_state()`, `set_webrelay_state()`.

Section 5

Safety Interface

5.1 Overview

The SHL subsystem implements several safety mechanisms to protect equipment and personnel. These include temperature monitoring with automated responses, smoke and fire detection, water intrusion detection, and power monitoring.

5.2 Temperature Monitoring

5.2.1 Temperature Thresholds

The system monitors shelter temperature and takes action based on the configurable thresholds listed in Table 5.1.

Condition	Default Threshold	Action
Normal	< 80°F	No action
Warning	80–90°F	Set status to WARNING
Critical	≥ 90°F	Set status to ERROR, shutdown critical ports

Table 5.1: Temperature Thresholds and Actions

Site-Specific Configuration

Threshold values are configurable per site via the `thermometers.warning_temp` and `thermometers.critical_temp` settings in `defaults.json`.

5.2.2 Critical Port Shutdown

When the temperature exceeds the critical threshold, the system can automatically power off designated "critical" PDU ports. The list of critical ports is defined in the configuration file under `thermometers.critical_list`.

Format: List of `[rack, port]` pairs.

Example:

```
"critical_list": [[1, 3], [1, 4], [2, 1]]
```

5.2.3 Temperature Status Messages

The system `INFO` field reflects the temperature status:

- `TEMPERATURE! Shelter temperature at XX.X F, shutting down critical ports: ...`
- `TEMPERATURE! Shelter temperature at XX.X F` (warning level)

5.3 Smoke and Fire Detection

New in Version F

Available through a correctly configured EnviroMux.

When an EnviroMux system is configured with a smoke detector, the SHL subsystem monitors for smoke/fire conditions.

5.3.1 Smoke Detection Response

When smoke is detected:

1. System status is set to `ERROR`
2. `INFO` is set to `SMOKE! Shelter smoke detector activated`
3. Event is logged

When smoke clears:

1. System status returns to `NORMAL`
2. `INFO` indicates error condition cleared

Note

The smoke detector response is informational only. No automated equipment shutdown is triggered by smoke detection. Human intervention is required for fire response.

5.4 Water Intrusion Detection

New in Version F

Available through a correctly configured EnviroMux.

When an EnviroMux system is configured with a water/leak sensor, the SHL subsystem monitors for water intrusion.

5.4.1 Water Detection Response

When water is detected:

1. System status is set to `ERROR`
2. `INFO` is set to `WATER! Shelter water sensor activated`
3. Event is logged

When water clears:

1. System status returns to `NORMAL`
2. `INFO` indicates error condition cleared

5.5 Door Monitoring

New in Version F

Available through a correctly configured EnviroMux.

When an EnviroMux system is configured with a door sensor, the SHL subsystem monitors the shelter door status.

5.5.1 Door Open Response

To avoid false alarms during maintenance, the door warning is delayed:

- Door open < 4 hours: No warning
- Door open ≥ 4 hours: Set status to `WARNING`, `INFO` indicates door is open

When the door is closed:

1. Warning condition is cleared
2. Door open timer is reset

5.6 Power Monitoring

New in Version F

Available through the LWA line voltage monitor board.

The voltage monitor detects power quality issues on the AC mains.

5.6.1 Power Flicker

A power flicker is a brief voltage drop that recovers quickly. When detected:

1. System status is set to `WARNING`
2. `INFO` includes `POWER-FLICKER! Flicker in the shelter power`
3. Flicker status clears automatically after 5 minutes

5.6.2 Power Outage

A power outage is a sustained loss of voltage. When detected:

1. System status is set to `ERROR`
2. `INFO` is set to `POWER-OUTAGE! Shelter power outage`
3. Outage state is saved to disk for persistence across restarts
4. When power is restored, status returns to `NORMAL`

Note

The voltage monitor tracks both 120V and 240V lines independently. Either line experiencing an issue will trigger the corresponding warning or error.

5.7 Device Unreachable Monitoring

The SHL subsystem monitors the reachability of all network devices (PDUs, thermometers, EnviroMux). When a device becomes unreachable:

1. Device is added to the unreachable list
2. If any devices are unreachable, status is set to `WARNING`
3. `INFO` shows count and list of unreachable devices

Devices are removed from the unreachable list after successful communication is restored (with hysteresis to prevent flapping).

5.8 Warning State Merging

Multiple warning conditions can exist simultaneously. The system merges warning states using the format:

```
{MIB1}! {MIB2}! ... {description1}; {description2}; ...
```

Example:

```
TEMPERATURE! DOOR! Shelter temperature at 82.0 F; Shelter door is open
```

When a warning condition clears:

- If other warnings remain: Update message to remove cleared condition
- If no warnings remain: Set status to `NORMAL`, set `INFO` to "Warning condition(s) cleared, system operating normally"

Section 6

References

6.1 LWA Documentation

1. MCS Common ICD (LWA Engineering Memo MCS0005)
2. MCS-NDP Interface Control Document
3. MCS-ASP Interface Control Document

6.2 External Software

lightning_detector Software for reading Boltek EFM-100C electric field mill data and broadcasting lightning strike notifications. Repository contains `spinningCan.py`.

URL: `https://github.com/lwa-project/lightning_detector`

voltage_monitor Software for monitoring AC line voltage using an Arduino-based sensor board and broadcasting power events. Repository contains `lineMonitor.py`.

URL: `https://github.com/lwa-project/voltage_monitor`

wview Weather station software that reads Davis VantagePro2 data and stores it in an SQLite database. The SHL subsystem reads from this database for weather MIB entries.

URL: `http://www.wviewweather.com/`

shelter The SHL subsystem software itself. Contains `shl_cmnd.py` (main command handler), `shlFunctions.py` (control functions), `shlThreads.py` (monitoring threads), `shlBard.py` (MC4002 interface), and `shlQube.py` (IceQube interface).

URL: `https://github.com/lwa-project/shelter`

6.3 Configuration Files

SHL configuration is stored in JSON format in site-specific files (`defaults.json.`*`site`*). The top-level keys and their nested structure are described below.

6.3.1 Global Parameters

The top-level scalar parameter identifies the SHL instance (Table 6.1).

Table 6.1: Top-level Configuration Parameters

Key	Type	Description
`serial_number`	string	SHL unit serial number (e.g., "SHL01")

6.3.2 `mcs` — MCS Communication

The `mcs` object configures the UDP interface between SHL and MCS (Table 6.2).

Table 6.2: `mcs` Configuration Parameters

Key	Type	Description
`message_host`	string	MCS host IP address
`message_out_port`	int	UDP port for sending responses to MCS
`message_in_port`	int	UDP port for receiving commands from MCS

6.3.3 `hvac` — HVAC Controller

The `hvac` object configures the HVAC controller interface and validation ranges for the `TMP` and `DIF` commands (Table 6.3).

Table 6.3: `hvac` Configuration Parameters

Key	Type	Description
`type`	string	HVAC controller type ("bard" or "ice-qube")
`temp_min`	float	Minimum allowed temperature set point (°F)
`temp_max`	float	Maximum allowed temperature set point (°F)
`diff_min`	float	Minimum allowed cooling differential (°F)
`diff_max`	float	Maximum allowed cooling differential (°F)
`ip`	array of string	HVAC controller IP addresses
`webrelay`	string	WebRelay IP address (IceQube type only)

6.3.4 `thermometers` — Temperature Sensors

The `thermometers` object configures standalone temperature sensor devices and the thresholds that trigger warning and critical actions (Table 6.4). The `devices` sub-object is a dictionary keyed by rack number.

Table 6.4: `thermometers` Configuration Parameters

Key	Type	Description
`warning_temp`	float	Warning temperature threshold (°F)
`critical_temp`	float	Critical temperature threshold (°F)
`critical_list`	array of array	`[rack, port]` pairs to shut down at critical temperature
`monitor_period`	float	Monitoring interval (seconds)
`thermometers.devices.`{*`id`*}		
`type`	string	Sensor type ("HWg" or "Comet")
`ip`	string	Device IP address
`port`	int	SNMP port
`security_model`	array	SNMP security: `[agent, community, version]`
`nsensor`	int	Number of sensors on the device
`description`	string	Human-readable description

6.3.5 `enviromux` — Environmental Monitoring

The `enviromux` object configures EnviroMux environmental sensor devices (Table 6.5). The `devices` sub-object is a dictionary keyed by device number.

Table 6.5: `enviromux` Configuration Parameters

Key	Type	Description
`warning_temp`	float	Warning temperature threshold (°F)
`critical_temp`	float	Critical temperature threshold (°F)
`critical_list`	array of array	`[rack, port]` pairs to shut down at critical temperature
`monitor_period`	float	Monitoring interval (seconds)
`enviromux.devices.`{*`id`*}		
`ip`	string	Device IP address
`port`	int	SNMP port
`security_model`	array	SNMP security: `[agent, community, version]`
`sensor_list`	array of string	Sensor types present (e.g., "door", "water", "smoke")
`description`	string	Human-readable description

6.3.6 `pdus` — PDU and UPS Devices

The `pdus` object configures power distribution units and uninterruptible power supplies (Table 6.6). The `devices` sub-object is a dictionary keyed by rack number.

6.3.7 `weather` — Weather Station

The `weather` object configures access to the wview weather database (Table 6.7).

Table 6.6: `pdus` Configuration Parameters

Key	Type	Description
`monitor_period`	float	Monitoring interval (seconds)
`pdus.devices.`{`id`}		
`type`	string	Device type ("APC", "Raritan", "Dominion", "TrippLite", "APCUPS", or "TrippLiteUPS")
`ip`	string	Device IP address
`port`	int	SNMP port
`security_model`	array	SNMP security: `[agent, community, version]`
`noutlet`	int	Number of controllable outlets
`description`	string	Human-readable description

Table 6.7: `weather` Configuration Parameters

Key	Type	Description
`database`	string	Path to wview SQLite database
`monitor_period`	float	Polling interval (seconds)

6.3.8 `lightning` — Lightning Alert Multicast

The `lightning` object configures the UDP multicast group for receiving lightning strike notifications from `spinningCan.py` (Table 6.8).

Table 6.8: `lightning` Configuration Parameters

Key	Type	Description
`ip`	string	Multicast group IP address
`port`	int	Multicast port

6.3.9 `outage` — Power Outage Alert Multicast

The `outage` object configures the UDP multicast group for receiving power flicker and outage notifications from `lineMonitor.py` (Table 6.9).

Table 6.9: `outage` Configuration Parameters

Key	Type	Description
`ip`	string	Multicast group IP address
`port`	int	Multicast port

Appendix A

Hardware Reference

This appendix provides a reference for the hardware supported by the SHL subsystem.

A.1 Command Exit Codes

Table A.1 lists the exit codes returned by SHL commands.

Code	Description
`0x00`	Process accepted without error
`0x01`	Invalid temperature set point
`0x02`	Invalid temperature differential
`0x03`	Invalid PDU rack number
`0x04`	Invalid PDU port number
`0x05`	Invalid PDU control keyword
`0x06`	Invalid command arguments
`0x07`	Blocking operation in progress
`0x08`	Subsystem already initialized
`0x09`	Subsystem needs to be initialized

Table A.1: Command Exit Codes

A.2 Subsystem Status Values

Table A.2 lists the possible values for the `SUMMARY` MIB entry.

A.3 Supported PDU Types

Type	Model	Outlets	Features
APC	APC AP7921B	8	Switching, current metering
Raritan	Raritan PX2 Series	8	Switching, V/I metering

Type	Model	Outlets	Features
Dominion	Raritan Dominion PX	Variable	Switching, V/I metering
TrippLite	PDUMH20ATNET	8	Switching, V/I/Hz metering

A.4 Supported UPS Types

Type	Model	Features
APCUPS	APC Smart-UPS	Battery status, charge level, output source, outlet switching
TrippLiteUPS	TrippLite SmartOnline	Battery status, charge level, output source, outlet switching

A.5 Supported Thermometers

Type	Model	Sensors	Notes
HWg	HW group STE	1–2	Supports external sensor probes
Comet	Comet P8510	1	Integrated sensor

A.6 Environmental Monitoring System

Type	Model	Supported Sensors
EnviroMux	Network Technologies E-MINI-LXO	Temperature (2), smoke, water, door, airflow

A.7 HVAC Controllers

Type	Model	Capabilities
bard	Bard MC4002	Multi-stage cooling/heating, lead/lag control, temperature sensors (local + 2 remote), alarms (fire, temp, controller)
iceqube	IceQube OUTDOOR	Single-stage cooling, lead status, temperatures (enclosure + condenser), filter status

A.8 Weather Station

Status	Description
`SHUTDWN`	Subsystem is shut down, monitoring threads stopped
`BOOTING`	Subsystem is initializing
`NORMAL`	Subsystem operating normally
`WARNING`	Warning condition exists (check INFO)
`ERROR`	Error condition exists (check INFO)

Table A.2: Subsystem Status Values

Component	Model	Data Provided
Weather Station	Davis VantagePro2	Temperature, humidity, pressure, wind, rainfall
Data Logger	WeatherLink	Serial/USB interface to wview software

A.9 Lightning Detection

Component	Model	Description
Electric Field Mill	Boltek EFM-100C	Measures atmospheric electric field, detects lightning strikes with distance estimation

A.10 Voltage Monitoring

Component	Model	Description
Monitor Board	LWA custom	Arduino-based board monitors 120V and 240V AC lines
Software	lineMonitor.py	Broadcasts power events via UDP multicast